

Polypress

A lossless compressor for data tables. Every number in this document was produced by the benchmark harness in this repository, on real public data, against production binaries — and this document is generated from that output rather than written from it.

500	478/500	1.25×	500/500
datasets measured	beaten, best of 17	median margin	round-trip exact

What it is

Polypress compresses tabular data — CSV, TSV, JSON, Parquet — and reproduces the logical table exactly: every column name, the column order, the row order, and every cell as the identical string. Every compression is verified in memory, cell by cell, before anything is written to disk.

The corpus, and why it is not chosen

The obvious objection to any compression result is *you picked the files*, and there is no way to answer it by picking more files. So the corpus is not picked. The Socrata open-data catalog — the index behind several hundred government portals — is asked for its datasets in descending order of lifetime page views, and the first 500 that survive purely mechanical filters are taken. Rank order is public and reproducible. Nothing is skipped for what is in it, and every rejection is recorded with its reason.

Total input: 3,572.9 MB. Sources: socrata500.jsonl.

The headline result

Codec	Total bytes	vs Polypress	Tables
polypress	222,835,247	1.00×	500
polypress+brotli	225,046,829	1.01×	500
polypress+zstd	241,691,904	1.08×	500
xz -9e	289,868,660	1.30×	500
brotli -q11	293,361,634	1.32×	500
zstd -22	308,874,691	1.39×	500
zstd -19	311,190,723	1.40×	500
parquet+brotli	315,314,683	1.42×	450
parquet+zstd	332,220,872	1.49×	450
bzip2 -9	352,197,023	1.58×	500
parquet+gzip	368,957,089	1.66×	450
orc+zstd	404,018,041	1.81×	436
orc+zlib	419,840,273	1.88×	436
zstd -3	490,402,314	2.20×	500

Codec	Total bytes	vs Polypress	Tables
parquet+snappy	510,369,991	2.29×	450
gzip -9	548,571,960	2.46×	500
orc+snappy	602,368,107	2.70×	436
lz4 -9	633,129,361	2.84×	500
feather+zstd	678,877,772	3.05×	450
feather+lz4	1,103,684,316	4.95×	450

Aggregated over the whole corpus Polypress is **1.26×** smaller than the best rival on each table taken **individually** — that is, against a competitor allowed to pick its best codec per dataset. Per table it wins **478 of 500**, median margin **1.25×**.

The "Tables" column is not decoration. Two competitors were silently missing from an earlier round of this benchmark because their writers raised on certain inputs and the exception was being swallowed. A codec measured on fewer tables than the rest is not comparable with the rest, so the count is printed.

Fidelity: what a size comparison hides

Parquet reproduced the exact printed text of the table on **81 of 437** datasets where the comparison could be made. On the rest it changed something — a number reformatted, a leading zero dropped, a date restyled. Those are real differences to anyone who has to hand the file back. **Its size column is therefore flattering on 356 of the 437 tables**, and it should never be quoted without this sentence next to it. Polypress restored every cell exactly on 500 of 500.

Is the margin the modelling, or just a better final compressor?

A fair objection, worth answering before it is asked. Parquet **cannot use xz at all** — its options are snappy, gzip, brotli, zstd and lz4, and asking pyarrow for xz returns an unsupported-codec error. So some of the margin above could be the finisher rather than the front end.

It is not. Re-finisher Polypress with the *same* codec Parquet is using:

Finisher	Polypress wins	Tables compared
zstd	449 of 450	450
brotli	449 of 450	450

Strip xz out entirely and the gap barely moves. The modelling is doing the work.

Why it wins

Three ideas. The third is the load-bearing one.

1. Predicting down a column

Fit a low-degree polynomial to the last few values in a column, extrapolate one step, store only the error. Because the fit is re-centred at every cell the coefficients never have to be stored — the decoder already holds the neighbours. Order- k extrapolation turns out to be exactly the k -th finite difference.

2. The same idea in two dimensions

Where adjacent numeric columns are *commensurable* — same decimal places, same magnitude — a cell is predicted from its left, upper and upper-left neighbours. Detection matters as much as the predictor: differencing "Model Year" against "Make" is meaningless, so groups form only where columns are genuinely comparable.

3. Cross-column structure by reordering rows

Table columns are not independent — City is nearly determined by Postal Code. Rather than model that, Polypress **sorts the rows by the parent column**: equal parents become adjacent, the child collapses into long runs, and the entropy coder eats it. **The permutation costs nothing to store**, because the decoder has already rebuilt the parent and recomputes the same stable sort. Parents are chosen by conditional entropy with a Miller-Madow correction and arranged into a tree.

This third idea is **prior art, not novel**: US 8,312,026 (Vo, AT&T, 2012, now expired) discloses it, down to the rule that a parent must be measured rather than assumed. It was arrived at independently here; the novelty claim is withdrawn.

Where it wins, and where it does not

Best eight:

Dataset	Input	Polypress	Best rival	Margin
data_weekly_united_states_covid_19	1.3 MB	34,434	orc+zstd 94,161	2.73×
data_breeding_bird_atlases_vk8g-yp	12.4 MB	97,581	parquet+brotli 261,155	2.68×
health_nursing_home_quality_initia	14.7 MB	112,835	xz -9e 301,108	2.67×
data_behavioral_risk_factor_survei	16.0 MB	230,624	bzip2 -9 596,917	2.59×
data_missouri_keg_tag_7fmu-y7e8.cs	6.9 MB	29,670	xz -9e 75,968	2.56×
data_alzheimer_s_disease_and_healt	16.0 MB	189,484	bzip2 -9 457,050	2.41×
data_nutrition_physical_activity_a	16.0 MB	161,437	xz -9e 383,108	2.37×
data_500_cities_local_data_for_bet	16.0 MB	357,969	xz -9e 831,688	2.32×

Every table it loses:

Dataset	Input	Polypress	Winner	Margin
data_sars_cov_2_variant_proportion	5.7 MB	368,268	orc+zstd 240,474	0.65×
data_energy_star_certified_smart_t	0.1 MB	15,177	brotli -q11 13,785	0.91×
data_open_meetings_au6r-w9n3.csv	0.1 MB	14,570	brotli -q11 13,473	0.92×
opendata_maryland_port_administrat	0.1 MB	18,552	brotli -q11 17,320	0.93×
data_new_york_state_budget_vetoes_	0.1 MB	18,843	brotli -q11 17,713	0.94×
data_missouri_river_water_trail_ac	0.1 MB	10,340	brotli -q11 9,853	0.95×
data_covid_19_vaccination_trends_i	10.3 MB	1,979,887	brotli -q11 1,905,037	0.96×
data_covid_19_vaccinations_in_the_	16.0 MB	2,778,869	brotli -q11 2,688,603	0.97×
data_salary_steps_by_job_classific	0.2 MB	21,762	brotli -q11 21,243	0.98×
data_2015_street_tree_census_tree_	16.0 MB	1,320,082	brotli -q11 1,290,579	0.98×
datos_listado_de_medicamentos_en_v	0.1 MB	15,711	bzip2 -9 15,456	0.98×
data_county_clerk_license_informat	0.6 MB	48,654	brotli -q11 48,174	0.99×
data_naloxone365_nj_free_naloxone_	0.1 MB	12,099	bzip2 -9 11,994	0.99×
datacatalog_medical_examiner_unide	0.1 MB	13,801	brotli -q11 13,700	0.99×
data_american_rescue_plan_arp_rura	3.8 MB	731,267	bzip2 -9 726,643	0.99×
data_energy_star_certified_residen	0.1 MB	17,492	brotli -q11 17,401	0.99×
data_tca_all_approved_grants_fy25_	0.4 MB	55,182	bzip2 -9 54,973	1.00×
opendata_csric_best_practices_qb45	0.7 MB	86,072	bzip2 -9 85,787	1.00×
data_tca_all_approved_grants_fy24_	0.3 MB	46,502	bzip2 -9 46,425	1.00×
data_current_sla_pending_licenses_	0.6 MB	107,813	bzip2 -9 107,692	1.00×
opendata_presubmission_community_m	0.4 MB	62,618	bzip2 -9 62,549	1.00×
data_provider_relief_fund_covid_19	0.1 MB	20,856	bzip2 -9 20,836	1.00×

Listed in full rather than summarised. A results document that reports only the wins is an advertisement.

Honest limitations

- **Speed.** Median encode 2.2 MB/s, decode 137 MB/s. Encode is the slow half and deliberately so: every table is also compressed by the plain fallbacks and the smaller result wins, which is what makes the never-worse guarantee true rather than assumed.
- **Encode cost grows with the square of the column count**, in the parent search. A very wide table is slow.
- **Genre.** Every dataset here is a government open-data table. That is a wide sample of one kind of data, not a sample of all data. Census microdata, scientific arrays, genomics and financial ticks are unmeasured, and the win should be expected to be smaller on some of them.
- **Nobody outside the project has run it.** Every number in this document was produced on one machine by one person.

Negative results

Kept because they are the most useful part of the record.

- **Run-length coding the text blob:** +0.3%. xz already subsumes it — RLE finds only adjacent repeats, LZMA finds them at any distance. The gain never comes from a better compressor on the same bytes; it comes from a better order.
- **Recovering ragged-decimal numeric columns:** 9–23% worse.
- **Raising the 50% dictionary threshold:** up to 2.45% worse, and it stayed worse after the theory that explained the failure was itself disproved.
- **Relaxing the two-dimensional exact-decimal rule:** zero bytes on every table.

Generated by docs/report.py from socrata500.jsonl. Reproduce with benchmarks/fetch_socrata100.py, benchmarks/sweep.py and benchmarks/report.py.